Miguel Xochicale

Driving scalable cloud robotics & AI infrastructure
for healthcare innovation.

Senior Research Software Engineer
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Education

Ph.D. in Computer Engineering at The University of Birmingham — Birmingham, UK
Nov 2014 – Nov 2018

Thesis: Nonlinear Analysis to Quantify Movement Variability in Human-Humanoid Interaction.
Supervisors: Professor Chris Baber and Professor Martin Russell
Awarded PhD degree: 12/07/2019. Thesis: GitHub: Website:

M.Sc. in Signal Processing at Institute of Astrophysics, Optics and Electronics (INAOE) — Puebla, México
Sep 2004 – Sep 2006

Thesis: Design of digital filters with fewer multipliers.
Supervisor: Dr. Gordana Jovanovic Dolecek
Awarded MSc degree: 01/11/2006. Thesis: GitHub:

Research Experience

Senior Research Software Engineer at University College London, UK — Oct 2022 – Present

- I lead development of scalable AI and cloud robotics infrastructure enabling UCL researchers to move from experimental models to deployable healthcare and cyber-physical systems. I build reproducible ROS2 workflows, APIs, and real-time sensor pipelines, strengthening institutional capability in robotics, digital health, and translational research engineering across multiple collaborative initiatives. (Cyber Physical Systems ; READY ; CDI-HUB ; SENSE-BASE ; SciKit-Surgery ; XFETUS NVIDIA-holoscan).

Research Associate in Real-time AI-based Ultrasound Imaging at King's College London, UK — Sep 2021 – Sep 2022

- PIs: Dr Andrew King and Dr. Alberto Gomez. I scientifically contributed to automatic biometric recognition of electrocardiography ultrasound data using real-time deep learning techniques with Python, CUDA, C++ and Qt programming languages via GitHub. See more at ( ).

Research Associate in Software and Hardware Engineering at King's College London, UK — Apr 2019 – Aug 2021

- PIs: Dr. Wenfeng Xia and Prof. Tom Vercauteren. I pushed forward the state-of-the-art of Ultrasound-Guidance Interventions, contributing to the development of a needle tip tracking system, real-time ultrasound image processing, quality management system for clinical translation of medical devices, and public engagement activities. See more at ( ).

Publications

1. H. Mannering, Y. Zhang, L. Ziao, H. Zhiwu, J. Matthew, and **M. Xochicale**, "A foundational edm2-based generative model for high-resolution synthetic fetal ultrasound imaging from open datasets," in *30th UK Conference on Medical Image Understanding and Analysis (**MIUA '26**)*, arXiv, Aug. 2026, 3 pages, 1 figure, appendix. code and data is open source.
2. B. Duvieusart, **M. Xochicale**, D. Kaski, and T. Leung, "Towards affordable smartphone eye tracking for nystagmus analysis and monitoring," *The 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025.
3. A. Das et al., "Automated surgical skill assessment in endoscopic pituitary surgery using real-time instrument tracking on a high-fidelity bench-top phantom," *Healthcare Technology Letters*, vol. 11, no. 6, pp. 336–344, 2024. eprint: `https://ietresearch.onlinelibrary.wiley.com/doi/pdf/10.1049/htl2.12101`.
4. P. T. H. Nhat et al., "Clinical benefit of ai-assisted lung ultrasound in a resource-limited intensive care unit," *Critical Care*, vol. 27, no. 1, p. 257, Jul. 2023.
5. S. Thompson, **M. Xochicale**, T. Dowrick, and M. Clarkson, "Using scikit-surgery for augmented reality in surgery research," *Hamlyn Symposium of Medical Robotics*, Jun. 2023.

See the full list of publications and associated links ( ).

Talks, Posters and Workshops

1. M. J Golomb, D. Guzman, A. Esterson, S. Ramos, and **M. Xochicale**, "Speed vs structure: Building reproducible ai workflows with the ucl unified ai research platform," Research Software Engineering Conference (RSE'2026) at University of Sheffield, UK, Sep. 2026, Presentation, Abstract.
2. **M. Xochicale**, "How open-source software is shapping the future of healthcare?," Guest lecture MSc MSHDS at University of Bristol, Mar. 2026, Presentation, Abstract.
3. **M. Xochicale**, "Modernizing ros 2 skills: Hacking and orchestrating cloud brains, physical sensors, and the network," Free and Open source Software Developers' European Meeting (FOSDEM), Jan. 2026, Presentation, Abstract.
4. **M. Xochicale**, I. Tseyzer, G. Carvalho, and S. Varakliotis, "From lab to launch: The evolving role of rses in driving sustainable and commercially viable digital solutions," University of Warwick, Sep. 2025, Talk.
5. **M. Xochicale**, "How open-source software is shaping the future of healthcare," Free and Open source Software Developers' European Meeting (FOSDEM), Feb. 2025, Presentation, Abstract.

See the full list and associated links ( ).

Supervision and Teaching Experience

University College London — Jan 2023 – Present

- Co-supervision: Kofi Quarshie and Mohamed Mahmoud, in2reserach summer placement *Project: Federated learning for eye segmentation.* ( ) *June 26' – Aug 26'*
- Co-supervision of PhD student Kelechi Imediegwu from the UCL Global Business School for Health *Passed Upgrade Viva March 2026* *Oct 25' – Sep 27'*
- Co-supervision of PhD student Elvira Cortese from the UCL Queen Square Institute of Neurology *Oct 24' – Sep 27'*
- Supervision: John Nocum, in2reserach summer placement *Project: Hacking real-time AI workflows for eye-tracking.* *June 25' – Aug 25'*
- Supervision: Sujon Hekim, in2reserach summer placement *Project: AI-based surgical skill assessment using transformer model.* *June 23' – Sep 23'*
- Supervision: Xiaoning Zhu, M.Sc in Health Data Science *Project: Automatic Medical Image Reporting with transformer-based models.* *Jan 23' – Aug 23'*
- Supervision: Qingyu Yang, M.Sc. in Health Data Science *Project: Fetal Brain Ultrasound Imaging synthesis with diffusion models.* *May 23' – Aug 23'*
- Technical Lead: CHME0039: Artificial Intelligence in Healthcare Group Project. Lecturers: Prof. Paul Taylor and Dr Kevin Tsang
  I led two group projects on *Fetal Brain Ultrasound Imaging Synthesis with Diffusion Models.* *Jan 23' – March 23'*
  I led two group projects on *End-to-End AI Workflow for Automated Multimodal Medical Image Reporting (AMMIR).* *Jan 25' – March 25'*

King's College London — Jan 2020 – September 2022

- Supervision: Pablo Prieto Roca and Samuel Eyob. *KURF projects on 3DGANs for fetal US imaging.* *Jun 2022 – Aug 2022*

- Supervision: Tsz Yan (Goosie) Leung, MSc in Medical Engineering and Physics. *Project: Simple US guidance intervention.* *Feb 2021 – Aug 2022*
- Supervision: Thea Bautista, M. Eng. in Biomedical Engineering. *Project: DCGANs for fetal US imaging.* *Oct 2021 – May 2022*
- Supervision: Guilherme Gomes de Figueiredo and Amal Hussein. *KURF projects on DCGANs for US imaging.* *Jun 2021 – Aug 2021*
- Supervision: Alexander Mitton, M.Sc. in Medical Engineering and Physics. *Project: Vibro-tactile stimulator for dystonia.* *Jan 2020 – Sep 2020*
- Teaching Associate: Medical Robotics. Lecturers: Dr. Christos Bergeles (Jan2020-Apr2022), Dr Hongbin Liu (Jan2021-Apr2021), Dr. Alejandro Granados (Jan2022-April2022) *Jan 2020 – Apr 2022*

The University of Birmingham Jun 2018 – Dec 2018

- Supervision: Dinghuang Zhang, M.Sc. in Computer Engineering. *Project: Tools for human-humanoid collaboration* *Aug 2018 – Dec 2018*
- Teaching associate: Engineering Maths 2. Lecturers: Prof. Martin Russell, Dr Carl Anthony *Aug 2017 – Dec 2017; Jan 2018 – Apr2018*
- Teaching Associate: Matlab Laboratories. Lecturer: Dr Edward Tarte *Jan 2017 – Apr 2017*
- Teaching Associate Computing for Engineering. Lecturer: Dr Sridhar Pammu *Aug 2016 – Dec 2016; Aug 2017 – Dec 2017*
- Teaching Associate: Small Embedded Systems. Lecturer: Prof. Chris Baber *Aug 2016 – Dec 2016*

Grants, Awards and Honours

- Small Acorns Fund by the The National Hospital and The Institute of Neurology for: *Real-time, AI-powered Diagnosis of Nystagmus*. Blog *(11/11/2026).*
- UCL Open Science award 2023 for organising "Open-Source Software for SurgTech" workshop , *(26/10/2023).*
- My M.Sc. student, Thea Bautista received the Maurice Wilkins Prize at KCL for the best MEng Individual Research Project in 2022 *(30/08/2022).*
- King's Public Engagement grant for the project "FETUS: Finding a fETus with an Ultrasound Simulator" led by myself and in collaboration with Fang-Yu Lin and Shu Wang *(07/01/2021 - 07/01/2022)*
- My M.Sc. student, Alexander Mitton, won the Outstanding Individual Project award *(15/10/2020)*
- King's Health Partners grant for the project "Sensory system abnormalities in childhood dystonia" lead by Verity McClelland and in collaboration with Carlos Seneci *(14/04/2020 - 9/06/2020)*
- "Towards Healthy Ageing with Humanoid Robots" was selected for a talk at the 2nd forum of Mexican Talent 2017 *11/01/2017*
- I won the best poster award at the XIV Symposium of Mexican Students in the UK at University of Edinburgh *16-18/06/2016*
- My project of a low-cost robot was selected among 125 applications received from 35 countries and presented at the 1st international public entrepreneurship program in Mexico (MECATE 2015). *20-24/07/2015*

Skills

Programming and software Python[2014-present], GNU/Linux Operating System user (OpenSuse, Debian & Ubuntu)[2005-present], R[2013-present], Robot Operating System (ROS)[2016-present], ROS2[2025-present], GNU-Octave (as well as MatLab)[2009-present], LaTeX[2006-present], C and C++[2015-present], Processing[2012-present], the shell[2010-present], GNU-emacs[2010-present], vim[2016-present], pandoc[2017-present], open-source enthusiast at GitHub (@mxochicale)[2015-present], continuous integration & continuous delivery [2019-present], CUDA Programming [2021-present], Docker, podman [2024-present], github container registires [2025-present], python-based linting [2022-present], aws tools [Oct2024-present], NVIDIA Isaac Sim, Isaac Lab and ROS2 [Jan2025-present], Kubernetes distributed trainer [Jan2026-present].

Tools and hardware Single-board computers and microcontrollers (NVIDIA Jetson Nano, RaspberryPi, BeagleBone, Arduino & PIC)[2010-present], Inertial Measurement Units (calibration, collection & data analysis)[2013-present], Web design (Github pages, Jekyll)[2015-present], Graphic design (Inkscape, GIMP)[2014-present], CAD design (Autodesk inventor, blender, FreeCAD, & onshape)[2015-present], Artificial Neural Networks (PyTorch, & TensorFlow)[2017-present], 3D printing (flsun, cura) [2019-present], Video framegrabbers (PCI & usb from ephipan), Clinical Ultrasound Devices (Voluson E10, Philips EPIC, convex & linear probes), Medical imaging (3D slicer, ITK-SNAP), NVIDIA Clara AGX with holoscan-sdk [2023-present], gtec EEG and g.USBamp [2025-present].

QMS Control documents with standard compliances using: Good Machine Learning Practice by FDA, Health Canada, & MHRA; IEC 62304 Medical Device Software Standard; BS EN 82304-1:2017 Health Software; IEC 60601-1 Medical electrical equipment; BS EN 61391-2:2010 Ultrasonics; BSI 60825-14 Safety of laser products.

Languages Spanish[Native], English[Fluent] and interested in learning Chinese.

Outreach activities and scientific engagement

- Lead organiser of the UCL workshop "Hacking Cyber-Physical Systems in Real Time," delivering hands-on training to 13 participants on real-time robotics pipelines using containerised ROS2 environments deployed on institutional infrastructure. GitHub: . *2 March 26'.*
- Lead organiser of "From Code to Care: Challenges, Success Stories, and the Future Directions of Open-Source development in Medical Robotics" at the Hamlym Symp. on Medical Robotics 2026. . *Jan 26' – Oct 26'.*
- Lead organiser of "Open-Source Software for Surg, BioMed and AI Tech 3" at the Hamlym Symp. on Medical Robotics 2025. . *Jan 25' – Oct 25'.*
- Lead organiser of HAWKESHACKS3 hackathon "Redefining Eye Tracking with Open-Source Real-Time AI Solutions": Site: , & GitHub: . *6-8th Nov 24'.*
- Lead organiser of "Open-Source Software for SurgTech 2" at the Hamlym Symp. on Medical Robotics 2024. . *Jan 24' – Oct 24'.*
- Lead organiser of "Open-Source Software for SurgTech" at the Hamlym Symp. on Medical Robotics 2023. GitHub: & Video Playlist . *Jan 23' – Oct 23'.*
- Lead organiser of CMICHACKS hackathon 2023 "Real-time AI for surgery": Site: , & GitHub repository: . *9-10 Nov 23'.*
- Co-organised of journal club for computer vision and deep learning at Advanced Research Computing Centre *01-01-2023 – 01-06-2023.*
- Co-organised of events at the BMEIS Early Career Researcher Network (monthly meetings & writing workshops). *01-01-2021 - 01-09-2022.*
- Participation in the Westminster Enterprise Week to engage students aged 14-18 to Biomedical Engineering. *10-11-2021*
- Participant in the STEAM WEEK organised by the City Westminster Council to engage students aged 14-18 to STEAM *23-03-2021*
- Alexandra Lautarescu and I organised the Reproducible, Interpretable, Open, & Transparent Science Club at BMEIS *02-2020 – 06-2020*
- For the event In2ScienceUK, I shared my scientific journey to young scientists on how they can become better scientist. *20-08-2019*
- For the New Scientist Live, I showcased software that helps doctors to create 3D models of brain tumors using AI. *09-2019*
- Finalist at the Three Minute Thesis Competition 2018. Video: and GitHub: *05-2018*
- PhD Research Poster Conference at Uni of Bham (2015) , (2016) , and (2018) . GitHub: . *2016 to 2018*
- Demoing Human-Robot Activities at the Undergraduate Open Days. GitHub: . *2014–2018*
- Building Artificial Intelligence and Robotics for Children. Twitter: @air4children GitHub: @air4chidlren *2019–Present*